INCH-POUND

MIL-DTL-85069/11A
25 January 2006
SUPERSEDING
MIL-S-85069/11
14 November 1986

DETAIL SPECIFICATION SHEET

SLEEVE (GROMMET), HARD SEAT, TENSION HEAD, SWAGED, COUNTERSUNK, PANEL FASTENER

This specification sheet is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-85069, "Sleeves, Fastener, General Specification for".

DESIGN INFORMATION: Hard seat sleeve grommets covered by this specification sheet are for use in the holes of aircraft removable composite aircraft panels.

REQUIREMENTS: The sleeve grommets shall be as specified herein and in accordance with figures 1, 2, and 3.

1. Material. The sleeve grommets shall be manufactured from A286 cres (UNS S66286) in accordance with SAE AMS 5525, "Steel, Corrosion and Heat Resistant, Sheet, Strip, and Plate 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V 1800 ºF (982 ºC) Solution Heat Treated".

2. Protective treatment. The sleeve grommets shall be passivated in accordance with SAE AMS-QQ-P-35, "Passivation Treatments for Corrosion-Resistant Steel".

3. Finish.

P - Cadmium plating in accordance with SAE AMS-QQ-P-416, "Plating, Cadmium (Electrodeposited)", for type II, class 2 as required.
N - No finish except passivation in accordance with SAE AMS-QQ-P-35.

4. Surface texture. Unless otherwise specified the surface texture shall be not more than 125 microinches in accordance with ASME B46.1, "Surface Texture (Surface Roughness, Waviness, and Lay)".

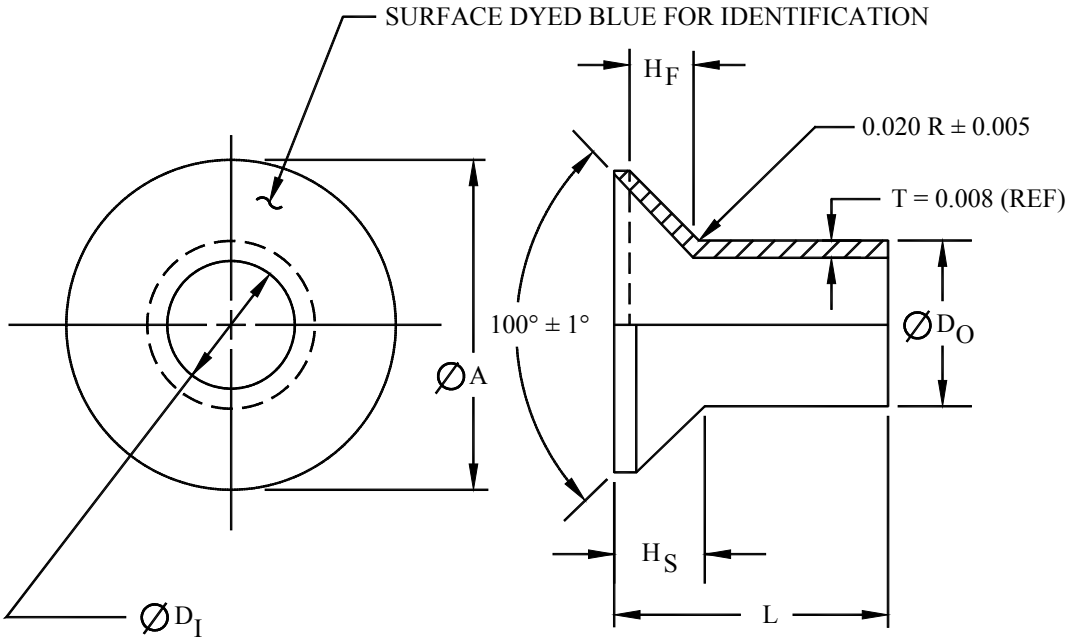
5. Dimensions. The sleeve grommets shall be manufactured to the dimensions listed in table I. Unless otherwise specified dimensions are in inches. Dimensions indicated are after plating.

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6. Tolerances. Tolerances are as indicated.

7. Lubrication. None.

8. Workmanship. Break all sharp edges and remove all burrs and slivers.



Dash no.	For fast size (ref)	ØA dia. ±0.005	D_I dia.	D_O dia.	H_f head height fastener[1]	H_s head height sleeve[1]	L ±0.010[2]	Recommend hole prep	
								Drill no.[3]	Drill dia.
08	1/4	0.505	0.2545 0.2530	0.2706 0.2696	0.108	0.113	1.000	--	0.273
10	5/16	0.631	0.3170 0.3155	0.3331 0.3321	0.135	0.140	1.000	--	0.334
12	3/8	0.758	0.3795 0.3780	0.3956 0.3946	0.162	0.167	1.000	--	0.398

[1]Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.
[2]Uncut sleeve length is 1.000 inch. Before installation cut sleeve length to equal panel thickness (see figure 2).
[3]Special drill is required.

FIGURE 1. Uninstalled grommet.

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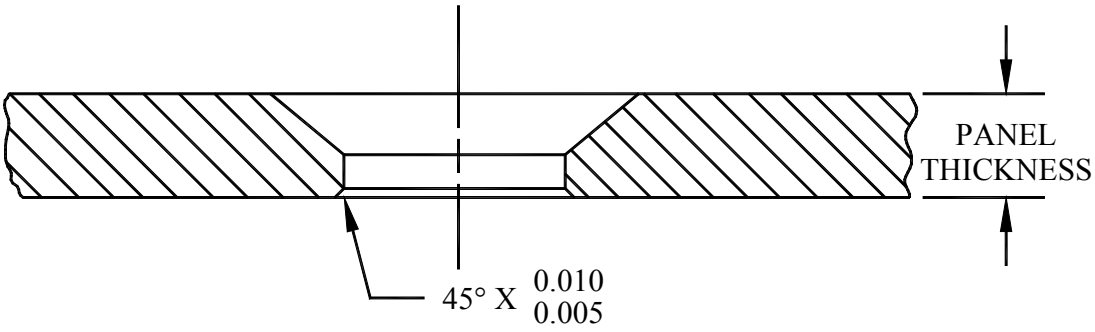


FIGURE 2. Grommet hole preparation.

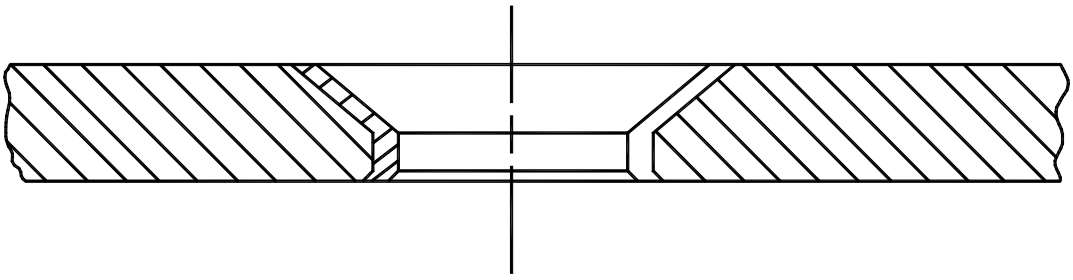


FIGURE 3. Installed grommet.

GENERAL INFORMATION:

9. Part or identification number (PIN). The PIN to be used for sleeve grommets acquired to this specification is created as follows:

M 85069/11-X XX

- XX — Fastener shank diameter dash number (see figure 1)
- X — Surface finish: N = none; P = cadmium plate, type II, class 2
- 85069/11 — Specification sheet number
- M — M prefix

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing Activity:
DLA - GS7

(Project 5365-0040)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil.

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